DIGITAL TIMELINE · EDUCATIONAL RESOURCE

THE EVOLUTION OF ARTIFICIAL INTELLIGENCE

From Turing's fundamental question to large language models — a comprehensive journey through the milestones, minds, and pivotal moments that built modern AI.

75+
YEARS OF HISTORY

2
AI WINTERS

∞
FUTURE POTENTIAL

TIMELINE OVERVIEW

1943–1956 FOUNDATION
The Foundations

1956–1974 BOOM
The Golden Age

1974–1980 WINTER
First AI Winter

1980–1987 BOOM
Expert Systems

1987–1993 WINTER
Second AI Winter

1993–2010 BOOM
Statistical Revolution

2010–2017 REVOLUTION
Deep Learning Revolution

2018–Now REVOLUTION
Foundation Models Era

01

THE FOUNDATIONS

1943 – 1956

1943

MP

NEURAL MODEL

McCulloch & Pitts Neuron

First mathematical model of a neuron — the conceptual ancestor of all modern neural networks.

1950

AT

LANDMARK PAPER

The Turing Test

"Can Machines Think?" — Alan Turing proposes the Imitation Game, asking the defining question of AI.

1956

JM

BIRTH OF AI

Dartmouth Conference

John McCarthy coins "Artificial Intelligence." A new scientific field is officially born.

02

The Golden Age

1956 – 1974

BOOM

1958

The Perceptron

Rosenblatt's early supervised learning algorithm — NYT predicted machines would "walk, talk, see."

1958

LISP Language

John McCarthy develops LISP — the dominant AI programming language for three decades.

1966

ELIZA Chatbot

MIT's first chatbot simulates a therapist. Users believe they're talking to a real person.

03

First AI Winter

1974 – 1980

WINTER

1969

Perceptrons Book

Minsky & Papert expose XOR limitations — nearly kills neural network research for a decade.

1973

The Lighthill Report

UK's damning review slashes AI funding. Combined with US DARPA cuts, a long funding drought begins.

1974

Funding Collapse

Both UK and US governments drastically reduce AI research investment. Progress halts.

1980

COMMERCIAL AI

Expert Systems Boom — XCON & R1

DEC deploys XCON, saving $40M/year. Hundreds of companies invest in expert systems. The AI industry hits $1 billion by 1985. Japan launches its $850M Fifth Generation Computer Project.

1986

Backpropagation — Neural Networks Revived

Geoffrey Hinton, Rumelhart & Williams popularize backpropagation for multi-layer networks. Solves the XOR problem and reignites connectionist AI after years in the cold.

05 — Second AI Winter · 1987–1993

AI Market Crashes (1987)

LISP machine hardware market collapses — cheap PCs outperform dedicated AI workstations. Expert systems prove brittle. DARPA cancels its Strategic Computing Initiative.

06

Statistical Revolution

1993 – 2010

1997 Deep Blue Defeats Garry Kasparov

HISTORIC MILESTONE

IBM's Deep Blue becomes the first computer to defeat a reigning World Chess Champion in a regulation match. Kasparov wins Game 1 but loses the series 2.5–3.5. Watched by millions worldwide, it marks a turning point in how society perceives machine intelligence.

1989

YL

Convolutional Neural Networks

LeCun's LeNet-5 at Bell Labs becomes the blueprint for modern image recognition — though the world won't notice for 20 more years.

1995

CV

Support Vector Machines

Powerful statistical classifiers become the dominant ML paradigm through the 2000s — spam filters, OCR, and early recommendation engines.

1998

GP

Google PageRank

AI-driven ranking transforms how humanity accesses information. A form of link analysis built on ideas from machine learning.

07

Deep Learning Revolution

2010 – 2017

2012 AlexNet — Deep Learning Wins ImageNet

IMAGENET MOMENT

Krizhevsky, Sutskever & Hinton's deep CNN achieves 15.3% top-5 error — nearly 11 points better than the runner-up. Trained on just two NVIDIA GPUs, it ignites a GPU arms race and every major tech company begins aggressively hiring deep learning researchers.

2014

IG

Generative Adversarial Networks

Ian Goodfellow invents GANs — two networks competing to generate realistic images. Foundational to modern generative AI.

2016

DM

AlphaGo vs. Lee Sedol

DeepMind's AlphaGo defeats world Go champion 4-1. Deep RL enters complex intuitive human domains once thought impossible.

2017

GV

"Attention Is All You Need"

Vaswani et al. introduce Transformers with self-attention — the architecture behind BERT, GPT, and every major language model.

2018

BERT & GPT — Pre-training Paradigm

Google's BERT and OpenAI's GPT prove large-scale text pre-training enables powerful transfer learning. The "pre-train then fine-tune" paradigm transforms NLP.

2021

DALL·E, CLIP & Stable Diffusion

OpenAI connects language and vision. Stable Diffusion democratizes AI image generation — millions generate photorealistic images from text for the first time.

2022

ChatGPT — AI Goes Mainstream

100 million users in 2 months — the fastest-growing consumer app in history. Built on GPT-3.5 with RLHF, it brings conversational AI to the general public and triggers a global race.

2023–25

GPT-4, Claude, Gemini, Llama

Multimodal models demonstrate unprecedented reasoning, coding & creativity. Models pass the bar exam, score 90th percentile on SAT. AI safety moves to the forefront of global policy.

AI's Impact Across Industries

How artificial intelligence is reshaping every sector of the global economy

HEALTHCARE

Diagnosis & Drug Discovery

AlphaFold solved the 50-year protein folding problem. AI detects cancers from imaging with radiologist-level accuracy and compresses drug development from years to months.

TRANSPORTATION

Autonomous Vehicles

Waymo, Tesla Autopilot, and Cruise deploy self-driving at scale using computer vision, sensor fusion, and deep reinforcement learning.

FINANCE

Trading & Fraud Detection

Over 70% of US stock trades are algorithmic. AI detects credit card fraud in milliseconds, saving banks billions through anomaly detection.

SCIENCE

Accelerating Discovery

AI assists climate modeling, materials science, genomics, and fundamental physics — compressing decades of research into years by exploring vast solution spaces.

THE STORY

Continues.

From a mathematician's thought experiment in 1950 to systems that pass the bar exam and assist Nobel Prize research — artificial intelligence has transformed from a philosophical curiosity into the defining technology of the 21st century.

KEY FIGURES

- AT — Alan Turing
- JM — John McCarthy
- GH — Geoffrey Hinton
- YL — Yann LeCun
- YB — Yoshua Bengio